TRIASHA SARKAR

Atlanta, GA · +1 (404) 948-8761 · tsarkar34@gatech.edu
linkedin.com/in/triasha-sarkar · github.com/triasha72 · triasha72.github.io/Portfolio

Machine learning engineer and Georgia Tech aerospace MS graduate who builds tested systems from data and evaluation through deployment. Experience spans search and ranking, recommendation, GenAI evaluation, time-series modeling, scientific ML, and safety-critical engineering.

EDUCATION

MS, Aerospace Engineering | Georgia Institute of Technology, Atlanta, GA Aug 2026
Graduate Research Assistant, Aerospace Systems Design Laboratory (ASDL)

MSc, Aerospace Vehicle Design | Cranfield University, United Kingdom Feb 2023
First Class Honours · Thesis on geometry optimization under FAA constraints

B.Tech, Aerospace Engineering | SRM Institute of Science and Technology, India Jun 2021
First Class Honours

TECHNICAL SKILLS

ML & GenAI: PyTorch, scikit-learn, Transformers, RAG, hybrid retrieval, reranking, recommender systems, uncertainty estimation, time-series ML
Systems: Python, Go, SQL, Kafka, FastAPI, Docker, Kubernetes, GitHub Actions, DuckDB, PostgreSQL/pgvector, Prometheus, ONNX, Core ML, Qualcomm QNN
Scientific computing: NumPy, SciPy, pandas, NetworkX, OSMnx, statistical inference, design of experiments, surrogate modeling

PROFESSIONAL EXPERIENCE

Graduate Research Assistant under Prof. Dimitri Mavris | Georgia Tech ASDL May 2025 – Aug 2026
- Research teams needed dependable evidence from simulation, safety, and sustainability studies, so I took each problem from data review and model design through tested analysis.
- I worked with faculty and sponsors to check assumptions, document limits, and deliver models that others could reproduce and use.

Machine Learning Engineer | ALTEN India, embedded at Rolls-Royce Oct 2023 – Apr 2025
- Lifecycle engineers needed clearer signals from multivariate engine data, so I built Python workflows for diagnostics, anomaly detection, predictive maintenance, and mixed-frequency time series. I reviewed each result with domain experts before it was used.
- I turned certification requirements into reproducible analyses and model checks, giving engineering teams a traceable basis for safety-critical decisions.

Data Science Intern | Rolls-Royce DataLabs May – Jul 2021
- I cleaned aircraft-engine sensor data, designed useful features, and compared predictive and anomaly-detection models. The analysis gave engineers a clearer view of recurring failure patterns.

TECHNICAL PROJECTS

AeroRAG-X | Retrieval and LLM Evaluation for Technical Knowledge 2025 – Present
github.com/triasha72/AeroRAG-X
- NASA technical reports are difficult to search and verify, so I built hybrid retrieval, reranking, pgvector search, and citation controls over 3,233 source-preserving chunks.
- I added protected evaluations, evidence checks, bounded agents, FastAPI, Docker, tracing, and a release gate that rejects placeholder GRPO results. This keeps unsupported answers and incomplete RL evidence from passing silently.

NewsLens | Recommendation, Real-Time Search, and Production ML Jul – Aug 2026
github.com/triasha72/NewsLens · Python, Go, Kafka, PostgreSQL, FastAPI, Kubernetes
- News recommendation can look stronger when future behavior leaks into training, so I used chronological splits, TF-IDF ranking, and a training-only fallback, reaching NDCG@10 of 0.366. I froze the design before a one-time MIND holdout; the change stayed positive, but its paired interval included zero.
- New articles needed a recoverable path into search, so I built a Go, Kafka, and PostgreSQL pipeline with idempotent consumers, dead letters, query intent, and freshness-aware ranking. A 500-event local run had no failures, 44 ms publish p99, 79 ms index-freshness p95, and recovered a stopped consumer's partition in 5.7 seconds.

EdgeGenBench | Scientific ML and On-Device Inference Aug 2026 – Present
github.com/triasha72/EdgeGenBench
- Aircraft design studies need fast predictions without losing physical checks, so I built a surrogate benchmark with uncertainty estimates, constrained optimization, ONNX export, and deployment tests.
- The Qualcomm QNN model retained 0.997 mean held-out R^2, while a drifting INT8 candidate was rejected by policy. I also shipped an installable Safari app for browser-based iPhone use without claiming native-device performance.

AeroSynth-Eval | Multimodal AI Evaluation Aug 2026 – Present
github.com/triasha72/AeroSynth-Eval
- Synthetic inspection benchmarks were easy to change during experimentation, so I built a fixed 48-scenario VLM suite with deterministic assets, provenance checks, annotation workflows, and typed evaluator contracts.
- A free Kaggle P100 batch attempted all 12 development cases; eight passed the response schema, and four JSON failures were kept for diagnosis. The run proves execution and reliability behavior, not human alignment.

Silent Failure Detection in Rocket-Motor Simulations May – Jul 2026

Georgia Tech ASDL · Graduate Research Assistant under Prof. Dimitri Mavris

- A design sweep was failing without errors or output. I audited the data and source code, built a leakage-safe classifier with 96.8% accuracy across 15,120 simulations, and traced the failures to an uncapped convergence loop.
- I tested the result on unseen geometries and a separate sampling study, then wrote a patch that turns silent hangs into logged convergence errors.

HERO | Source-Aware Retrieval for Airline Safety 2025 – 2026

Georgia Tech ASDL · Graduate Research Assistant under Prof. Dimitri Mavris

- For Delta's HERO program, I worked on retrieval and evaluation methods that keep generated safety answers tied to the technical sources analysts need to review.

GREEN TEA and Project EAGLE | Sustainable Aviation Modeling Sep 2025 – Apr 2026

Georgia Tech ASDL · Graduate Research Assistant under Prof. Dimitri Mavris

- I built surrogate, uncertainty, demand, and life-cycle models for alternative-fuel and transport studies. The GREEN TEA model is still used by the sponsor.

Atlanta Mobility Resilience Digital Twin 2026 – Present

github.com/triasha72/atlanta-mobility-resilience-digital-twin

- I built an OSMnx and NetworkX simulator to test how road closures change travel time and access across Atlanta. Ruff and deterministic core tests now run in CI, while live OSM download remains an explicit integration step.

Surrogate Model Learning 2025 – Present

github.com/triasha72/Surrogate-model-learning

- I compared Gaussian process, response-surface, and radial-basis models to see where each method breaks down. All three notebooks now run from clean kernels in CI, so the results do not depend on hidden notebook state.

Equity Backtest | Walk-Forward Signal Evaluation Jul 2026

github.com/triasha72/Equity-Backtest

- I built an expanding-window backtest with transaction costs and kept every specification tested. The apparent momentum edge disappeared after costs and did not meet the significance threshold set before the analysis.

LEADERSHIP

Graduate Chair | Women of Aeronautics and Astronautics, Georgia Tech 2025 – 2026